

ABL Customer Statement Intelligence Suite

Agentic AI System — Statement Q&A, Subscription Analysis & ABL Policy Q&A

Document Type	Software Requirements & Design Specification
Prepared By	Daniyal Saqib
Role	IT Intern (ABIP), Allied Bank Limited
Sponsor / Reviewer	Sir Affan Wahid
Project Duration	27 Aug 2026 - 25 Sep 2026 (29 days)
Application	App 2 of 2
Status	Draft v1.1

Modules Covered in This Document

- 1 Account Statement Q&A Agent
- 2 Statement Subscription / Recurring Payment Analysis Agent
- 3 ABL Policy Q&A Agent (retrieval-augmented, built on public ABL documents)

This specification is written to be implementation-agnostic: it defines what each module must do, how modules interact, and what data and behaviour are required, without prescribing a specific programming language, framework, or hosting choice. Technology selection is a separate decision left to the implementing engineer.

Table of Contents

1	Introduction
2	System Architecture
3	Data Model
4	Interface Design
5	Non-Functional Requirements
6	Risks & Mitigations
7	Success Criteria
8	Project Timeline
9	Assumptions
10	Glossary of Terms

1. Introduction

1.1 Purpose

This document specifies an agentic AI system that lets a customer or new employee ask natural-language questions about a bank statement or about Allied Bank’s publicly published products and policies, and receive grounded, explainable answers. All statement data used is synthetic; all policy content is drawn from ABL's public-facing materials only. The system is designed to show that an agent can answer accurately and cite its source, rather than producing confident but ungrounded answers.

1.2 Stakeholders

Role	Interest
Sir Affan Wahid	Internship supervisor and primary reviewer.
Daniyal Saqib	Sole developer; responsible for design, implementation, and demonstration.
Prospective end customer (simulated)	The persona the Statement Q&A and Subscription Analysis agents are designed for.
Prospective new employee (simulated)	The persona the ABL Policy Q&A agent is designed for.

1.3 Scope

In scope for this delivery:

- Natural-language Q&A over a single uploaded/mock account statement
- Automated detection and summarization of recurring payments/subscriptions within a statement
- Retrieval-augmented Q&A over ABL's publicly available product and policy documents (e.g. ABIP program page, account types, published fee schedules)
- A shared retrieval and reasoning layer used by all three modules

Explicitly out of scope:

- Any connection to a real customer's live statement or account
- Any non-public ABL internal document or system
- Financial or legal advice — outputs are informational summaries only, with a clear disclaimer in the UI

1.4 Data Constraint

Allied Bank Limited does not, and will not, provide real customer, transaction, or account data for this project. All data used for development, testing, and demonstration is synthetically generated and stored locally. No production ABL system is connected to, queried, or modified at any point. Any resemblance to real account numbers, balances, or customer names in sample data is coincidental and clearly marked as mock data throughout the build and demo. This constraint applies uniformly across every module described in this document and is treated as a hard requirement, not a preference.

2. System Architecture

Two data paths feed the system: (1) a customer's own statement, parsed on upload and never persisted beyond the session, and (2) a pre-indexed corpus of ABL's public documents used only for the Policy Q&A module. A router classifies each incoming question and dispatches it to the correct agent, which is restricted to its own tools and data source — the Policy agent can never see statement data and vice versa.

Layer	Responsibility
Presentation Layer	Conversational interface used for both statement Q&A and policy Q&A.
Query Router	Classifies each incoming question as statement-related or policy-related and dispatches it to the correct agent.
Statement Q&A Agent	Answers questions against a parsed statement using structured lookups.
Subscription Analysis Agent	Detects recurring transaction patterns within a statement and summarizes them.
ABL Policy Q&A Agent	Retrieves relevant passages from an indexed ABL public-document corpus and answers with citations.
Retrieval Store	Holds searchable representations of ABL's public documents for the Policy Q&A Agent.
Data Layer	Stores parsed statement line items and detected subscriptions for the active session only.

2.1 High-Level Data Flow

- 1 A statement is uploaded, or a policy question is asked, by the user.
- 2 The Query Router classifies the request and dispatches it to the correct agent.
- 3 For a statement query, the relevant agent reads only the parsed line items for the active session.
- 4 For a policy query, the Policy Q&A Agent retrieves the most relevant passages from the indexed corpus and answers with a citation to the source document.
- 5 The response, along with any citation, is returned to the presentation layer for display.

2.2 Design Principles

- **Separation of concerns** — each agent owns exactly one responsibility and never reaches into another agent's data directly.
- **Everything is logged** — no agent action is allowed to happen silently; every decision produces an entry in the audit trail.
- **Fail closed, not open** — when an agent is uncertain, the system defaults to the safest, most conservative outcome rather than guessing.

3. Data Model (Mock / Synthetic)

All entities below are populated with generated data at seed time. No entity is ever sourced from a live ABL system. Field names describe intent, not a fixed schema — the implementing engineer may adapt types and storage as appropriate for their chosen approach.

StatementLine

Represents one parsed line item from an uploaded statement, scoped to a single session.

Field	Notes
line_id	Primary key
session_id	Ties this line to one upload session; not persisted beyond the session
date	Transaction date
description	Line item description as it appeared on the statement
amount	Decimal value
category	Agent-assigned spending category

DetectedSubscription

Represents a recurring payment pattern identified within a session's statement lines.

Field	Notes
subscription_id	Primary key
session_id	Ties this record to one upload session
merchant	Detected merchant or payee name
frequency	One of: weekly / monthly / annual
avg_amount	Average amount across detected occurrences
last_seen	Date of the most recent occurrence

PolicyDocumentChunk

Represents one indexed passage from ABL's public documents, used only by the Policy Q&A Agent.

Field	Notes
chunk_id	Primary key
source_url	The public ABL page the chunk was drawn from
content	The extracted passage text
embedding	A vector representation of the passage, used for similarity search

4. Interface Design

The interfaces below describe the actions each module must expose, independent of how they are ultimately implemented (REST, RPC, direct function calls, etc.).

Action	Description
Upload statement	Parse an uploaded statement into line items for the active session.
Ask about statement	Ask a natural-language question about the uploaded statement.
List subscriptions	List detected recurring payments for the active session.
Ask about policy	Ask a natural-language question against the indexed ABL public-document corpus.

Action	Description
Reindex policy corpus	Re-crawl and re-embed the public document corpus (admin only).

5. Non-Functional Requirements

Category	Requirement
Data retention	Uploaded statement data must not persist beyond the active session — deleted on session end.
Groundedness	The Policy Q&A Agent must cite the specific source document/section for every answer; it must not answer from general knowledge if retrieval returns nothing relevant.
Isolation	The Policy Q&A Agent's retrieval index must contain only public ABL documents — no internal or synthetic-customer data may be indexed alongside it.
Accuracy disclaimer	All statement summaries must display a visible 'informational only, not financial advice' notice.
Freshness	The Policy Q&A Agent must display when its document corpus was last indexed, so answers are never presented as more current than they are.

6. Risks & Mitigations

Risk	Likelihood	Mitigation
Policy corpus goes stale if ABL updates public pages	Medium	Timestamp the index and show 'last indexed on' in the UI; support manual reindex.
Statement parser fails on inconsistent layouts	Medium	Support a simple structured-file upload as a reliable fallback alongside document parsing.
Subscription detection false positives (irregular but non-recurring spend)	Low	Require at least 3 occurrences at a consistent interval before classifying as a subscription.
Policy agent answers confidently despite weak retrieval match	Medium	Enforce a minimum relevance threshold; below it, the agent must say it cannot find a grounded answer.

7. Success Criteria

The following criteria define what a working demonstration must show by the end of the project window. These are the minimum bar for the build to be considered complete, independent of any stretch goals.

#	Criterion
1	A demo statement can be uploaded and answered against correctly in a live session.
2	At least one genuinely recurring payment in the demo data is correctly detected and summarized.
3	Every policy answer shown in the demo includes a visible citation to its source document.
4	A policy question with no good match in the corpus is met with an honest 'cannot find this' response, not a fabricated answer.

8. Project Timeline (27 Aug - 25 Sep 2026)

Week	Dates	Milestone
Week 1	27 Aug - 02 Sep	Statement parser + line-item model; mock statement generator
Week 2	03 Sep - 09 Sep	Statement Q&A Agent + Subscription Analysis Agent complete
Week 3	10 Sep - 16 Sep	ABL public-document corpus collection, indexing pipeline, Policy Q&A Agent
Week 4	17 Sep - 23 Sep	Chat interface integration, end-to-end testing
Buffer	24 Sep - 25 Sep	Polish, demo rehearsal, documentation finalization

9. Assumptions

- ABL's public website content is stable enough during the project window that the indexed corpus does not need continuous re-crawling.
- Sample statements can be generated with enough realism (varied merchants, believable recurring charges) to make the Subscription Analysis Agent's output meaningfully demonstrable.
- 'New employee' and 'customer' personas can share the same Policy Q&A Agent, since both are asking questions against the same public-document corpus.
- Sir Affan's review is a milestone check-in, not a formal client acceptance process — scope may be adjusted after each week based on feedback.

10. Glossary of Terms

Term	Definition
Agent	A component that autonomously reasons over a task, may call tools, and produces a structured decision or output.
Retrieval-augmented answering	Answering a question by first retrieving relevant source passages, then generating a response grounded in them, with a citation.
Session	The lifetime of one statement upload and the questions asked against it; data does not outlive the session.
Corpus	The full set of indexed public ABL documents available to the Policy Q&A Agent.
Groundedness	The property of an answer being directly traceable to a specific retrieved source, rather than generated from general knowledge.